

PARAMETER REGISTER AND REVIEW RISKS

Fixed numbers, their sources, and what a reviewer will not accept

Kashmir bus route rationalisation → *Transport Policy* manuscript

Assembled 2026-10-08 · plan version 3.4.5-geo

How to use this document

Part 1 is the short answer. Part 2 ranks the fixed numbers by how much of the plan they decide. Part 3 is the full register: every fixed number, where it came from, what it changes, and what would make it credible. Part 4 lists what will not survive a serious review, with the reviewer's likely sentence and our best response. Part 5 describes the ways the paper can fail. Part 6 is the work list.

Each number carries one of six grades.

Grade	Meaning
Standard	A published manual, specification or study gives this value or a range containing it.
Regulator	Set in review with the transport regulator. A policy choice.
Operator	Taken from the e-bus operator's data or design.
Fitted	Tuned so the model reproduces the e-bus totals.
Judgement	Chosen by the authors. No source.
Unsupported	The code names a source that no project file holds.

Sources marked **(checked)** were opened and read during the source research of 5–6 October. Sources marked **(to check)** are quoted from memory of the standard literature. Nobody on the project has opened them for this purpose, and each must be read before it is cited in the paper.

Part 1 — the short answer

- 1 **Most of the fixed numbers are the authors' judgement.** A handful sit inside a published range, the headway rules were set with the regulator, a few come from the e-bus operator, one is fitted, two cite a source the project does not hold, and one is wrong. Part 3 grades each.
- 2 **The destination weights (hospital 1.0, school 0.4, tourist site 0.6) have no source, and they do not matter.** Changing them, re-tiering whole categories, or weighting every destination equally moves at most 4 of 186 routes to another class.
- 3 **The larger problem with destinations is which tier each one was put in.** Every mapped place of worship was scored like a major hospital: 547 of the 1,030 top-tier destinations. This was not in the earlier parameter table. It also does not change the classes (at most 4 routes).
- 4 **The destination layer adds almost nothing to the index.** With destinations removed altogether, 173 of 186 routes keep their class. The "composite" index is close to a population-density ranking. A reviewer can use this against us ("why have it?") and we can use it for us ("the judgement weights cannot have driven the result").
- 5 **Three sets of numbers decide the fleet, and none is derived.** The headway rules (15, 20, 35 and 50 minutes), the cap on minutes per kilometre (which fixes the cycle time on 169 of 186 routes), and the 15 % spare allowance. About half the fleet (503 of 1,011 buses) comes from the headway rules, not from estimated demand.
- 6 **The "demand-responsive" rural rule is in practice a 50-minute rule.** Of 67 rural routes outside the e-bus network, 62 sit at the 50-minute maximum and 5 at 35. None landed on 40 or 45.
- 7 **No independent check passes.** The building-footprint check is circular, the e-bus benchmark fails and is circular, the expert panel was not run, and the GPS check fails on travel time.

- 8 **The paper is still worth submitting if it argues what the evidence supports.** Four findings hold: 614 permits are 157 corridors; straight-line catchments overstate reach; the speed cap, not the congestion model, sets the fleet; and the sensitivity ranking says which data to collect first. The paper fails if it claims a validated demand-based plan.

Part 2 — which numbers decide the plan

Rank	Number	What it decides	Measured effect	Basis
1	Headway rules: 15 / 20 / 35 / 50 min	Fleet on every route	Buses by headway: 283 at 15 min, 120 at 20, 303 at 35, 305 at 50. Sizing by estimated demand instead gives 551 buses	Regulator and operator
2	Cap on cycle time: 4.0 / 2.5 / 1.5 min per km	Cycle time on 169 of 186 routes	With the cap removed the same model gives 1,648 buses; at observed pace 1,130–1,266	Judgement
3	Which permits merge (65 % overlap, start points within 2.5 km, and only into a cluster's lead route)	The route set itself	Among the 186 surviving routes, 207 pairs still pass the merge test with each other. Merging connected groups at a 50 % threshold leaves 84 routes and 533 buses	Judgement
4	Spare allowance: 15 %	Fleet total	985–1,059 over 5–25 %; 94 % of the variance of the fleet as specified	Inside a published range
5	Walking distance: 400 m	Coverage	20.3 % at 300 m to 34.5 % at 800 m; 97 % of the variance of coverage	Convention
6	Population-to-destination split: 50 : 50	Route class	20:80 to 80:20 moves at most 18 routes; 93 % of the variance of class stability	Judgement
7	Overrides on class (e-bus lock, social floor, district-headquarters floor)	Published class of about 44 routes	Published classes match the index-only classes on 68.3 % of routes (79.5 % without the e-bus routes)	Judgement and operator
8	Destination weights and tiers	Route class	At most 4 routes	Judgement
9	Congestion multipliers, dwell, bridge delay	Cycle time on the 17 routes the cap does not bind	Fleet 999–1,016 across their ranges	Judgement; one unsupported

Everything below rank 4 changes little. A reviewer who reads Part 3 will still ask about each one, because the paper presents them as a method.

Part 3 — the register

A. Destination scoring

The engine adds the weights of all destinations within 250 m of a route, divides by route length and scales the result from 0 to 1 across routes.

Number	Value	Grade	Where it came from	Effect	To make it credible
Tier 1 weight	1.0	Judgement	Reference level. 1,030 destinations	—	Expert survey (below)
Tier 2 weight	0.4	Judgement. The code's reason, "education drives about 30 % of women's ridership", is Unsupported : the e-bus data has no trip purpose	Raised from 0.2 in the Jammu version of the engine. 1,080 destinations	0.2–0.6: no route changes class	Drop the stated reason. Expert survey

Number	Value	Grade	Where it came from	Effect	To make it credible
Tier 3 (seasonal) weight	0.6	Judgement	321 destinations. No visitor counts were used	0.3–0.9: 1 route changes class	Scale by published visitor numbers per site, or set equal to Tier 2
Winter weights	0.0 tourist-only; 0.4 others	Judgement	Winter switch is off in the published plan	Not used	Say that no winter plan exists
Weight if tier missing	0.4	Judgement	All 2,431 destinations carry a tier, so this never applies	None	State
Boost for women-oriented destinations	× 1.25	Judgement, motivated by the 64.5 % women's share of e-bus boardings	Applies to 12 destinations (6 girls' schools, 4 maternity hospitals, 1 girls' college, 1 anganwadi)	Not tested; too few to matter	Remove it. It adds an undefended number for no effect
Search distance	250 m	Judgement	Shorter than the 400 m used for residents; no reason recorded	Varied with stop spacing: up to 3 routes	Use the same distance as residents, or justify the difference

How destinations were put into tiers. This is a second, larger set of judgements. The extraction script first matches about 70 name patterns (SKIMS, Lal Ded, Hazratbal, Lal Chowk and so on), then falls back on the OpenStreetMap tag.

Rule in the script	Result in the data	Problem
Any place of worship → Tier 1, labelled "jamia_masjid"	547 points, 53 % of Tier 1. 401 are not named Jamia or Jama. The label is applied to all religions; one sampled row is a temple	A neighbourhood mosque counts the same as SKIMS and 2.5 times a school
Shop tagged mall or department store → Tier 1	138 points, including small shops and office complexes	Kashmir does not have 138 malls
Any hospital tag → Tier 1	155 points, including primary health centres and one under construction	A primary health centre equals a tertiary hospital
Hotel, guest house or hostel → Tier 3 (0.6)	240 points, including student hostels	A hotel outweighs a school (0.4)
Park → Tier 2	244 points	As many parks as schools (245)
Pharmacy → Tier 2 "small medical"	80 points	A pharmacy equals a college
University tag → Tier 2 unless the name matches four listed universities	47 colleges	Depends on spelling in OpenStreetMap

Two thirds of all destinations are in Srinagar district, so the layer mostly measures how thoroughly volunteers have mapped a place. Four routes have no destination within 250 m at all.

Test run for this document (analysis/checks/poi_retier_check.py; index-only classes, the engine's own 250 m rule; the baseline reproduces the analysis pipeline on 186 of 186 routes):

Change	Routes changing class	Agreement
Places of worship not named Jamia/Jama: 1.0 → 0.4	1	99.5 %
Same, dropped	2	98.9 %

Change	Routes changing class	Agreement
All places of worship dropped	4	97.8 %
Worship, shops and hotels all at 0.4	2	98.9 %
Worship and hotels dropped, shops at 0.4	4	97.8 %
Every destination weighted 1	3	98.4 %
No destination layer (population only)	13	93.0 %

These counts are for the index alone. The published classes also carry the overrides in group C.

Can a source be found for such weights? No manual gives per-category trip-attraction weights of this kind. Three defensible routes exist:

- 1 *Elicit them.* A pairwise-comparison survey (the analytic hierarchy process, Saaty 1980 **(to check)**) with 5–8 people: regulator's officers, the e-bus operator, Prof. Kathuria. Report the consistency ratio and the plan's classes under the elicited weights. About 20 minutes per person.
- 2 *Derive them from data.* Trip-generation rates by land use exist for Indian cities in comprehensive mobility plans; the Srinagar plan by RITES would be the natural source if the regulator can supply it. The e-bus boardings by stop, if the operator can supply them, would allow a regression of boardings on nearby destinations. Neither is in the project today.
- 3 *Stop defending them.* Report the table above and say the layer is a minor adjustment to a density ranking. This costs nothing and is the honest position.

B. Population and catchment

Number	Value	Grade	Where it came from	Effect	To make it credible
Walking distance to a route	400 m	Convention. The UN indicator for access to public transport uses 500 m (checked) . Daniels and Mulley (2013) find walking distance depends on supply (checked) . The usual citation, the <i>Transit Capacity and Quality of Service Manual</i> , could not be opened (to check) . India's 2009 benchmarks appear to set no walking distance (to check)	—	Coverage 20.3 % (300 m) to 34.5 % (800 m)	Cite the UN indicator and Daniels and Mulley. Report 400 m and 500 m side by side
Sampling interval along a route	250 m	Judgement (a computing choice)	—	Coverage 23.3–24.9 % over 150–400 m	Report as a numerical setting
Off-network tolerance in the walking catchment	100 m (one raster cell)	Judgement, paper's analysis only	Swept 50–150 m	Part of the 28.7–52.8 % range in straight-line overstatement	State; the range is already reported
Cap on population density before scaling	95th percentile	Judgement	—	Not tested	Add to the sweep (90th–100th)

Number	Value	Grade	Where it came from	Effect	To make it credible	
Tourist-corridor multiplier	× 1.3	Judgement	Applied to the per-route resident count of 8 active routes	1.0-1.8: no route changes class	Remove. Null effect, undefended number	
Ceiling on a route's resident count	2,000,000	Judgement	A guard against errors	None	State	
Population surface	WorldPop R2025A v1, constrained, 100 m	Standard (dataset). It is an alpha release and "may change" (checked)		—	Denominator 6,584,762. District totals are 0.71-1.20 of the 2011 Census	Say "alpha" and give the access date. Report district coverage with that error

C. Index and route class

Number	Value	Grade	Where it came from	Effect	To make it credible
Population : destination split	50 : 50	Judgement	Equal weighting is the usual default discussed in the OECD handbook on composite indicators (to check)	20:80 to 80:20: at most 18 routes change class	Report with the entropy weights (computed) and elicited weights
Scaling	Minimum–maximum	Standard method	—	Sensitive to outliers; partly handled by the 95th-percentile cap	State
Number of classes	3, fixed in the engine	Judgement	The paper's rule "smallest number reaching a fit of 0.80" has no citation. Two classes score 0.786, three 0.902	A threshold of 0.78 would give two classes	Say three classes were a design choice that the fit statistic supports; do not present the 0.80 rule as a derivation
Road-role multiplier	Trunk 1.25, feeder 0.75, other 1.00	Judgement	Used only as a tie-break within 5 % of a class boundary	Not tested	Add to the sweep, or remove
Bonus for routes connecting to the e-bus network	× 1.5	Judgement	Applied when choosing the lead route of a cluster	Not tested	Add to the sweep
Gate for promotion to trunk	30th percentile of the index	Judgement, lowered from the 50th	Computed over all 644 rows, including duplicates	Not tested	Add to the sweep
Minimum trunk length	5 km	Judgement	—	Small	State

Number	Value	Grade	Where it came from	Effect	To make it credible
E-bus routes locked to the top class	30 routes	Operator	20 of them fall in the lowest index class	Largest single cause of the 68.3 % agreement	Report both 68.3 % and 79.5 %
Social floor: lowest class raised to middle	11 hand-listed places (4 migrant townships, 3 Srinagar hospitals, 4 district hospitals), 250 m	Judgement. The district hospitals of Baramulla, Bandipora, Kupwara, Kulgam and Shopian are not listed	37 routes flagged	Not tested	List the places in the supplement; add the five missing district hospitals or explain
Route type by length	under 15 km urban; 15–40 km peri-urban; over 40 km regional	Judgement. The permit's vehicle category can override it	Decides the cap, the minimum fleet and the headway rule	Not tested	Replace with district and tehsil boundaries already in the project, or add to the sweep

D. Merging permits

Number	Value	Grade	Effect	To make it credible
Buffer around route lines for overlap	80 m	Judgement	Tested with the threshold	State
Share of length that must overlap	65 %	Judgement	0.50–0.90 swept for pair counts only: 354 to 90 qualifying pairs among surviving routes, 207 at 65 %	Report the route count and fleet at each value
Start points within	2,500 m	Judgement	Removes about two thirds of overlapping pairs (614 pass on overlap alone at 65 %, 207 with this condition). Never varied	Add to the sweep
Merge only into the cluster's lead route	A rule, not a number	Judgement	Large. It is why 207 qualifying pairs remain unmerged. Merging connected groups at 50 % gives 84 routes and 533 buses, an upper bound with fleets not re-sized	Explain the rule in §4.6 and report the scenario beside it. This is the least-examined large choice in the plan
Corridor identity (endpoints hashed)	11 m	Judgement	Gives 157 corridors	Report the count at 50 m and 100 m
Name-matching threshold for e-bus routes	0.80, or one shared distinctive word	Judgement	Earlier value 0.45 wrongly matched 11 routes as e-bus routes; corrected before the published plan	None needed

E. Travel time and cycle time

Number	Value	Grade	Where it came from	Effect	To make it credible
Congestion multiplier, city core	× 2.2	Unsupported. The code cites "Srinagar Traffic Police bottleneck studies"; no such file exists	Raised from 1.4 after an internal audit	Fleet 1,003–1,014 over 1.6–2.8	Never repeat the cited source. Replace with GPS speeds
Congestion multiplier, elsewhere	× 1.4	Judgement	—	As above	Same
"City core"	Every point north of latitude 34.07	Judgement, and wrong as geography	170 of 186 routes are treated as city core, including routes in Kupwara and Baramulla	Hidden by the cap	Disclose. A reviewer who finds this unaided will distrust the rest

Number	Value	Grade	Where it came from	Effect	To make it credible
Stop spacing for time	500 m	Inside the usual 400–600 m range (to check)	—	Fleet 1,001–1,016 over 300–800 m	Cite after checking
Time lost per stop	0.5 min	Low. Observed in driver GPS: about 0.87 min per assumed stop	The code's own description says 1.5	Fleet 999–1,016 over 0.3–1.5	Use the observed value
Terminal layover	10 % of running time	Inside the 10–15 % textbook range (to check)	Reduced from 15 %	Not tested	Cite after checking; add to the sweep
Penalty per sharp turn	0.5 min above 75°	Judgement	—	Small	State
Jhelum crossing delay	8 min, multiplied by the congestion factor	Judgement. The crossing test is a single line of longitude	—	Not tested	Measure from GPS runs that cross the river
Cap on cycle time per km	Urban 4.0, peri-urban 2.5, regional 1.5 min/km one way	Judgement. The code calls it a "sanity cap"	Implies at least 15, 24 and 40 km/h	Binds on 169 of 186 routes. Observed urban pace is 4.62 min/km at the median and 6.28 at the 90th percentile	See Part 4, item 2
Fallback speeds and distance factors	10 / 18 / 28 km/h; 1.25; 1.60 across the river	Judgement	Used only if the routing engine is down	Not used in the published run	State that they were not used

F. Headway and fleet

Number	Value	Grade	Where it came from	Effect	To make it credible
Headway, e-bus routes	15 min	Operator's stated design target. No document held	—	283 buses	Obtain the operator's document
Headway, other high-class routes	20 min	Regulator. The code also cites Bengaluru (10–15 min) and Mysuru (20–30 min) with no source	Was 15	120 buses	A written note from the regulator
Headway, other city routes	35 min	Regulator ("one hour is too long")	Was 30 and 60	303 buses	Same
City maximum	35 min	Regulator	—	—	Same
Rural steps	35, 40, 45, 50 min	Regulator and lead author ("waits must not exceed about 50 minutes")	Earlier steps ran to 120 min	62 of 67 routes sit at 50; none at 40 or 45	Describe it as a 50-minute maximum wait. Do not call it demand-responsive
Target load on rural routes	55 %	Judgement	—	Has no effect at the 50-minute maximum	Say so

Number	Value	Grade	Where it came from	Effect	To make it credible
Headway by permit category	"city bus" at least 20; "mps" at least 35; "regular" 35	Judgement	Legacy from earlier versions	Not disclosed in the paper	Disclose or show it changes nothing
Spare allowance	15 %	Inside the usual 10–20 % range (to check) . The code's "operator's target" has no document	India's road-transport undertakings had 87.6 % of buses on the road in 2019–20 and 72.3 % in 2021–22 (checked) , which implies 14 % and 38 %	Fleet 985–1,059 over 5–25 %	Cite the ministry's utilisation figures; they are Indian and already verified
Minimum buses per route	2 urban and peri-urban; 1 regional	Judgement	—	Small	State
Maximum buses per route	45	Judgement	—	Rarely binds	State
E-bus fleet floor	Operator's deployed buses	Operator	Binds on 2 routes	With the five GPS-timed routes, the difference between 1,000 and 1,011	Disclosed
Service day	16 h in the plan; 11 h (13 h e-bus) in timetables	Judgement; timetable day from observed hours	—	Cost and trips	Settled

G. Vehicle mix

Number	Value	Grade	Effect	To make it credible
Routes under 12 km	No full-size buses	Regulator and judgement	With the operator's own mix on e-bus routes, 25 of 32 trunks have none	State the rule as it is
Routes of 12 km or more	Full-size buses at most 50 %	Regulator ("neither class a majority"); was 85 %, then 60 %	Medium buses are the majority on 29 of 32 trunks	Already corrected in §4
E-bus routes	Split scaled from the operator's 9 m and 12 m counts	Operator	Not disclosed	One sentence
Capacity	60 / 35 / 20 passengers	Near India's Urban Bus Specifications (to check) ; the 20 is judgement. The code calls 60 a "crush load"	Load factors only	Cite after checking

H. The ridership estimate (Eq. 8)

It affects load factors, the rural headway rule and the demand-sized fleet. It does not affect routes or classes.

Number	Value	Grade	Where it came from	To make it credible
Trips per person per day	1.6	Judgement; Indian city studies report lower values for cities of this size (to check)	Chosen with the mode share so the product matched the e-bus totals	Use a cited value; the fitted scale absorbs the change

Number	Value	Grade	Where it came from	To make it credible
Bus share of trips	9 % urban; 7.2 % peri-urban; 5.4 % regional	Judgement. The 0.8 and 0.6 steps have no source	—	Cite a mode-share table, or state as assumption
Scale	0.33	Fitted	11,632,326 e-bus boardings in May 2025–April 2026 ÷ 365 = 31,869 a day. Was 0.18 before the routes were re-geocoded	Fit on half the routes or months and test on the rest
Peak-hour share	10.67 % of daily boardings	Operator data, April 2026	One month, one operator, city routes only	State the scope

The three numbers above multiply together, so only their product is identified. The estimate is fitted and tested on the same 30 routes. At route level it does not rank the e-bus routes correctly (rank correlation 0.22, $p = 0.24$).

I. Engine columns the paper does not use

Fare ₹10; operating cost ₹65 per km; subsidy threshold 0.6; long-journey flag 45 min; diesel 950 g CO₂ per km; e-bus 30 g CO₂ per km (wrong by a factor of about 26 against grid-based figures); winter walking factor 0.65; winter service ratio 0.85. All judgement. One sentence in the supplement should say the paper recomputes cost and emissions and ignores these columns.

J. Settings in the paper's own analysis

Setting	Value	Basis	Issue
Ranges and distributions in the uncertainty analysis	12 inputs, uniform or triangular around the plan's value	Judgement	The output ranges are only as good as these choices. The dwell range was widened upward after seeing the GPS
Draws; seed	5,000; 20260823	—	Fine
Pass marks for the GPS check	error under 20 %, rank correlation over 0.5	Judgement, written in the same commit as the first results	Cannot be called pre-registered
Pass mark for the e-bus benchmark	± 15 %	Judgement; the engine's own check used ± 25 %	Same
Pass mark for the building check	rank correlation over 0.60	Judgement; the building layer was changed after the first one failed	Disclosed
Plausible growth band for population check	0.5–3.0 % a year	Judgement	8 of 10 districts fall outside it
"Frequent" service	15 min	Convention	Fine
Funded share in the staged-funding result	30 %	Judgement	Report the whole curve
Target loads in the demand-sized fleet	50 %, 70 %, 85 %	Judgement around textbook values	Fine as a range
Diesel economy, full-size bus	3.5–5.0 km per litre	Named source not held. The ministry's review gives 1.95–5.66 for city undertakings (checked)	Replace with the checked range

Setting	Value	Basis	Issue
Diesel economy, minibus; all costs per km	7-10 km per litre; ₹25-90 per km	Judgement	Cost results stay "provisional"
E-bus energy, 12 m	0.98-1.3 kWh per km	ITDP India 2022 (checked)	Fine. No 9 m figure exists
Grid emission factor	0.675-0.705 kg CO ₂ per kWh	Central Electricity Authority, version 22 (checked)	Fine
Kilometres per bus per day, national	218.2	Ministry review, 2021-22 (checked)	Fine

K. Fixed inputs that are data, not choices

Input	Value	Caveat
Permit register	614 records, from the regulator; 679 valid bus permits on 14 March 2026	Not open data. The reduction of the regulator's 1,467 rows to these 614 is not described in the paper
E-bus aggregates	30 routes, 98 buses, 11.63 million boardings, May 2025-April 2026	Permission to publish not yet in writing
Driver GPS	2,526 runs, about 157 devices, February-June 2026	Consent screen added after collection. Around Srinagar only; two rural corridors
Road and footpath map	OpenStreetMap, northern-India extract of 6 January 2026	Completeness uneven outside Srinagar
Destinations	2,431 from OpenStreetMap	Two thirds in one district
Today's fleet	777 = 679 + 98	A valid permit is not proof a bus runs

L. Hand-made inputs that behave like parameters

These are not numbers in a configuration block, so earlier tables missed them. Each changes the plan.

Input	What was done	Effect
Road distance on 49 routes	Replaced by values a language model researched from the web. No person checked them. The record keeps no source links	Fleet on those routes 247 → 207
Village locations	37 villages no geocoder could find were placed at their district's centre	19 routes, 93 buses
Cycle time on 5 routes	Re-timed from driver GPS	Fleet 1,004 → 1,011. The same five routes are then used in the GPS check
The 30 e-bus routes	Typed in by hand, without intermediate points	Geometry and length of the routes carrying 283 buses
Route lines on 18 routes	Redrawn after the distance changes; 3 could not be reconciled	Catchments on those routes
Tourist zones	19 hand-entered coordinates with 2 km and 0.6 km buffers	Which 8 routes get the × 1.3
Version history	About 15 versions under regulator review; fleet between 855 and 1,144	The plan is a negotiated outcome as much as a computed one

Part 4 — what will not survive review

Ordered by damage. "Cheap" means days of analysis with data already held.

1. The headways are not a result, and they produce half the fleet

Reviewer: "The frequencies are the regulator's preferences. By the authors' own estimate the median route would run about one-ninth full at peak. What is the method contributing?"

Why they are right. 508 of 1,011 buses are explained by estimated demand at a 70 % load target; 503 by the service standards. Median estimated peak load is 0.113, and 168 of 186 routes are below 0.40. On the e-bus routes boardings would have to grow 2.2 to 4.5 times.

What we can do. Nothing analytical makes these numbers derived. Present them as service standards chosen by the regulator, which is ordinary coverage-oriented planning (Walker's ridership-versus-coverage distinction). Put the demand-sized fleet of 551 beside 1,011 in the abstract. Obtain one written line from the regulator confirming the standards. Cite a level-of-service scale for headway **(to check)** so that 15 to 50 minutes can be placed on a published scale.

2. The speed cap is the model

Reviewer: "Equation 11 is decoration. On 91 % of routes the fleet is route length times a constant divided by headway, and the constant has no source and is faster than the buses you measured."

Why they are right. The cap binds on 169 routes. It implies 15 km/h in the city; the observed median is 13 km/h and the slowest tenth is under 10 km/h. Rural pace was observed on two corridors only.

What we can do (cheap). Derive the urban and peri-urban caps from the GPS itself (for example the 75th percentile of observed pace by class) and report the fleet under them. State the rural cap as untested. Lead with the observed-pace fleet (1,130–1,266) wherever 1,011 appears.

3. Nothing independent validates the plan

Reviewer: "Four validation channels: one circular, one failed and circular, one not run, one failed."

Why they are right. WorldPop R2025A uses Microsoft building footprints as an input **(checked)**, so agreement with them is expected. The e-bus benchmark tests the plan against the data it was tuned to and fails the ± 15 % band. Travel time fails in sample (25.6 % error on 5 corridors) and out of sample (31.1 % on 8). Route length passes on 5 corridors.

What we can do. Stop using the word "validation" for the footprint and e-bus checks; call them consistency checks. The one channel that could be independent is the August field observation, and that needs the raw sheets. A week of boarding counts at 10–15 stops would be worth more than any further modelling.

4. The index has no shown link to demand, and its second component is nearly inert

Reviewer: "A composite of two components correlated at 0.88, with arbitrary weights, never tested against ridership. Why should its order mean anything?"

Why they are right. The only route-level test available gives a rank correlation of 0.22. Removing the destination layer changes 13 routes. The tier assignment of destinations is crude (group A).

What we can do (cheap). Report the re-tiering table. Describe the index as a density ranking with a destination adjustment. Remove the women's boost and the tourist multiplier, which add undefended numbers for no effect. Run the expert survey if the regulator's officers are willing.

5. The plan cannot be regenerated, and parts of it were set by hand

Reviewer: "The released code produces 1,044 buses, not 1,011. Forty-nine distances came from a language model with no sources."

What we can do. The provenance paragraph in §4.2 already discloses this. Add the fleet with the original routed distances on those 49 routes as a sensitivity row (about 1,050 by simple addition of the 40 buses removed; it has not been computed properly). Ship the correction logs. Say plainly which results are reproducible (the analysis from frozen inputs) and which are not (the plan itself).

6. "Robust" is claimed for the wrong object

Reviewer: "97 % stability is the stability of your re-implementation's classes. The published classes agree with it on 68 %."

What we can do. Already corrected in the numbers sheet. The abstract, §6 and §8 must lead with 68.3 % and 79.5 %, and explain that the gap is the e-bus lock.

7. The uncertainty ranges are not uncertainty about the fleet

Reviewer: "The spare allowance explains 94 % of the variance because the cap freezes everything else. The route set, the cap, the headway rules and the number of classes are not sampled."

What we can do (cheap). Add the cap values, the merge conditions and the headway of the middle class as scenario rows. Call 989–1,058 "the range under the plan's own rules".

8. The coverage headline depends on settings with no ground truth

Coverage is 24.2 % at 400 m, 20.3 % at 300 m and 34.5 % at 800 m. The straight-line overstatement ranges from 28.7 % to 52.8 % across settings. The population surface is an alpha release with district errors up to 29 %. *Response:* report the ranges in the abstract, not a single figure.

9. Smaller points a careful reviewer will raise

- The "city core" is a line of latitude covering 170 of 186 routes.
- The 2.5 km merge condition and the lead-route rule are never varied, although the route set depends on them.
- The 30th-percentile trunk gate is computed over rows that include duplicates.
- Five district hospitals are missing from the social-floor list.
- The benchmark "44 buses per lakh" is a think-tank's blend, not a ministry figure **(checked)**.
- The peer-city regression explains 10 % of the variance and cannot discriminate any fleet.
- The 13,087 residents "losing access" is a proxy with a range of 3,822–13,087.
- All cost figures rest on unverified unit rates.

10. Matters outside the analysis that can end the paper first

Ethics and consent for the GPS; written permission for the e-bus data; the declaration of AI use, which must now also cover the AI-researched distances; Srinagar-only scope and old counts in §1, §3 and §7; length.

Part 5 — how the paper can fail

Outcome	What triggers it	Likelihood as the draft stands	What removes the trigger
Rejected by the editor without review	Over length; blank ethics paragraph; section 1 contradicting the abstract on scope and counts; prose that reads as machine-drafted; "a local case study"	Real	Finish Part E of the decisions list; state the general contribution in the first paragraph
Rejected after review	A reviewer concludes the method is unsourced rules agreed with a regulator, with no successful validation, described as a demand-free planning method	Real if the framing stays	Reframe as an audit of an open-data plan: what such a plan can and cannot establish
Major revision	Reviewers accept the framing and ask for derived caps, an out-of-sample demand check, elicited weights and field counts	The best realistic first outcome	Do the cheap items in Part 6 before submission so the revision is small
Accepted, then challenged	Someone re-runs the public repository and gets 1,044, or finds mosques weighted as hospitals	Avoidable	Disclose both in the paper and the repository before anyone else does

The claims that hold up, in the order a reviewer would rank them:

- 1 614 permits describe 157 corridors, and a 71 % "route reduction" is almost entirely a change of unit.
- 2 Straight-line catchments overstate reach by 29–53 % depending on settings; coverage falls from 35.5 % to 24.2 % at 400 m.
- 3 A per-kilometre cap, added as a safeguard, became the rule that sets the fleet. Measured against driver GPS it is too fast, and the fleet at observed pace is 12–25 % higher.
- 4 The plan's fleet is half demand and half service standard, and the paper can say which half is which.
- 5 Sensitivity analysis ranks what to measure first: bus speeds, then walking distance, then demand.

The claims that do not hold up: a demand-based or demand-validated plan; a validated plan of any kind; a robust published hierarchy; an index that reflects demand; transferability shown by one case.

Part 6 — work list

Can be done now with data already held

#	Task	What it buys
1	Publish this register as the supplement's parameter table, with grades	Declared judgement is criticised far less than hidden judgement
2	Add the re-tiering table to §5 and describe the index as a density ranking with a destination adjustment	Closes the weights question
3	Derive urban and peri-urban caps from GPS and report the fleet under them	Replaces the weakest large number with a measured one
4	Fit the ridership scale on half the e-bus routes and test on the rest; repeat by month	Turns "fitted" into "fitted and tested", or shows honestly that it fails
5	Add to the sweep: the merge conditions, trunk gate, road multiplier, connection bonus, layover, bridge delay, population cap, route-type thresholds	Every untested number becomes a tested one
6	Fleet with the original routed distances on the 49 desk-checked routes	Bounds the effect of the least defensible input

#	Task	What it buys
7	Remove from the described method: women's boost, tourist multiplier, the "30 % of women's ridership" and "Traffic Police" statements	Fewer undefended numbers
8	Replace the diesel economy range with the ministry's checked figures	One fewer unverified cost constant

Needs a person

#	Task	Who
9	A written line confirming the headway standards were set in review	Regulator, through Avny ma'am
10	The operator's design headway, actual spare ratio, and whether a trip is counted one way	E-bus operator
11	Raw sheets of the August field observations	Avny ma'am
12	Pairwise-comparison survey for destination weights, 5–8 respondents	Lead author to circulate
13	A week of boarding counts at 10–15 stops, or stop-level e-bus boardings	Regulator or operator
14	Open and check every source marked "to check"	Any co-author with library access

Sources

Checked on 5-6 October 2026 (details and links in the research notes): UN indicator 11.2.1 metadata; Daniels and Mulley (2013), *Journal of Transport and Land Use* 6(2); Ministry of Urban Development (2009), *Service Level Benchmarks for Urban Transport*, bands confirmed through ITDP India (2022); ITDP India (2022), *Guidance for Electric Bus Rollout in Indian Cities*; Central Electricity Authority (2026), CO₂ baseline database, version 22; Ministry of Road Transport and Highways, *Review of the Performance of State Road Transport Undertakings 2019–20 to 2021–22*; WorldPop Global2 release statement R2025A; Microsoft Global ML Building Footprints licence; Centre for Science and Environment and CITIES Forum (2026), *Reinventing the Urban Bus*.

To check before citing: Transportation Research Board (2013), *Transit Capacity and Quality of Service Manual*, 3rd edition (walking distance, stop spacing, dwell, headway level of service); Vuchic (2005), *Urban Transit: Operations, Planning and Economics* (layover, spare ratio); Ceder (2007), *Public Transit Planning and Operation* (layover); Ministry of Urban Development (2013), *Urban Bus Specifications II* (capacities); Ministry of Urban Development (2008), study on traffic and transportation policies in urban India (trip rates, mode shares); Saaty (1980), *The Analytic Hierarchy Process*; Walker (2012), *Human Transit* (ridership and coverage goals); OECD (2008), *Handbook on Constructing Composite Indicators* (already in the reference list).